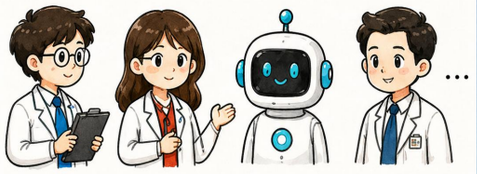
Data Collection

Diverse LLM Doctor Agents



MentalHospital



Stage-level Interaction Data

- Subjective Interview (S)
- Objective Examination (O)
- Diagnostic Assessment (A)
- Treatment Planning (P)

Stage 1 Rubric-grounded SFT

Expert Consensus Rubric



LLM Judge Ensemble (5 Judges)



Evaluation Trajectories & Scores (1–5)



Consensus Filter

High-quality Trajectories



SFT Cold Start



Low-confidence Pool



Stage 2 Expert-guided DPO

Low-confidence Pool



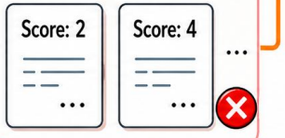
Matching Trajectory Selection



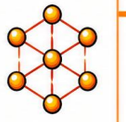
Chosen (same score)



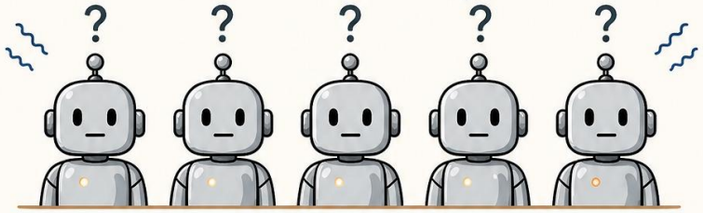
Rejected (other scores)



DPO Alignment



General LLM Judges



inconsistent scoring



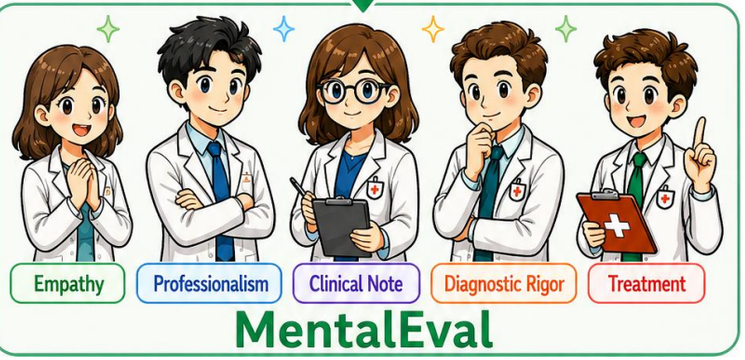
Stage 1: Rubric-grounded SFT

Learn from rubric criteria and exemplar rationales



Stage 2: Expert-guided DPO

Align with expert preferences through comparative feedback



MentalEval